

Statement of Work

AI-Powered Auto Claims Assessment Platform

Version: 1.0 — Prototype + Production Roadmap

Date: 2026-04-28

1. Project Scope

Vision

Reduce time-to-estimate, improve consistency, and lower handling cost on auto-physical-damage claims by automating the first-pass damage assessment with a vision-grounded AI pipeline. Replace the manual desk-review step on simple claims with an AI assessment that adjusters trust, and tier higher-stakes claims into human review automatically.

In scope (Phase 1 — Production v1)

- Image-based damage assessment. Web and mobile-web upload of a single damaged-vehicle photo (or image URL); AI returns vehicle metadata (make, model, color, year), damage description with severity classification, and a repair-cost range with cited sources.
- Confidence-driven routing. Per-field numerical confidence indicators; automatic flagging of low-confidence and high-severity claims for human adjuster review.
- Auditability. Every analysis is persisted with the input image, output JSON, model version, and external sources cited. Reanalysis on demand.
- PII handling. Server-side EXIF stripping and image normalization before upload to AI provider; license-plate and face redaction pipeline.
- Carrier integration. REST/webhook integration with the carrier's claims management system (Guidewire ClaimCenter, Duck Creek Claims, or equivalent).

Out of scope (Phase 1)

- Multi-angle / multi-image fusion (Phase 2)
- Pre-repair vs post-repair comparison flows (Phase 2)
- Fraud-signal pipeline (Phase 3)
- Native mobile apps (web-first; PWA-capable)
- Total-loss decisioning (advisory only — final call stays with adjuster)
- Subrogation, salvage value, or actual cash value calculations (out-of-band integrations)

Boundaries and assumptions

- Advisory, not authoritative. AI output is a recommendation. A licensed adjuster owns every claim decision, especially payouts and denials.
- Pilot cohort. Phase 1 deploys to a defined cohort (e.g., one product line, one geography). National rollout is a separate engagement.
- Carrier provides: branded UI assets, sandbox claims system access, sample claim photos and ground-truth assessments for evaluation, and an SME contact for adjuster workflow questions.

2. Technical Approach

Core architecture

A single, vision-capable LLM (Anthropic Claude Sonnet 4.6, with Opus 4.7 cross-validation on high-stakes claims) drives the AI pipeline. The prototype demonstrates that one model API call can produce all three required outputs — vehicle metadata, damage assessment, and repair cost — with structured-output guarantees. Production extends this with persistence, PII redaction, carrier integration, and a separate batch workload for non-latency-sensitive operations.

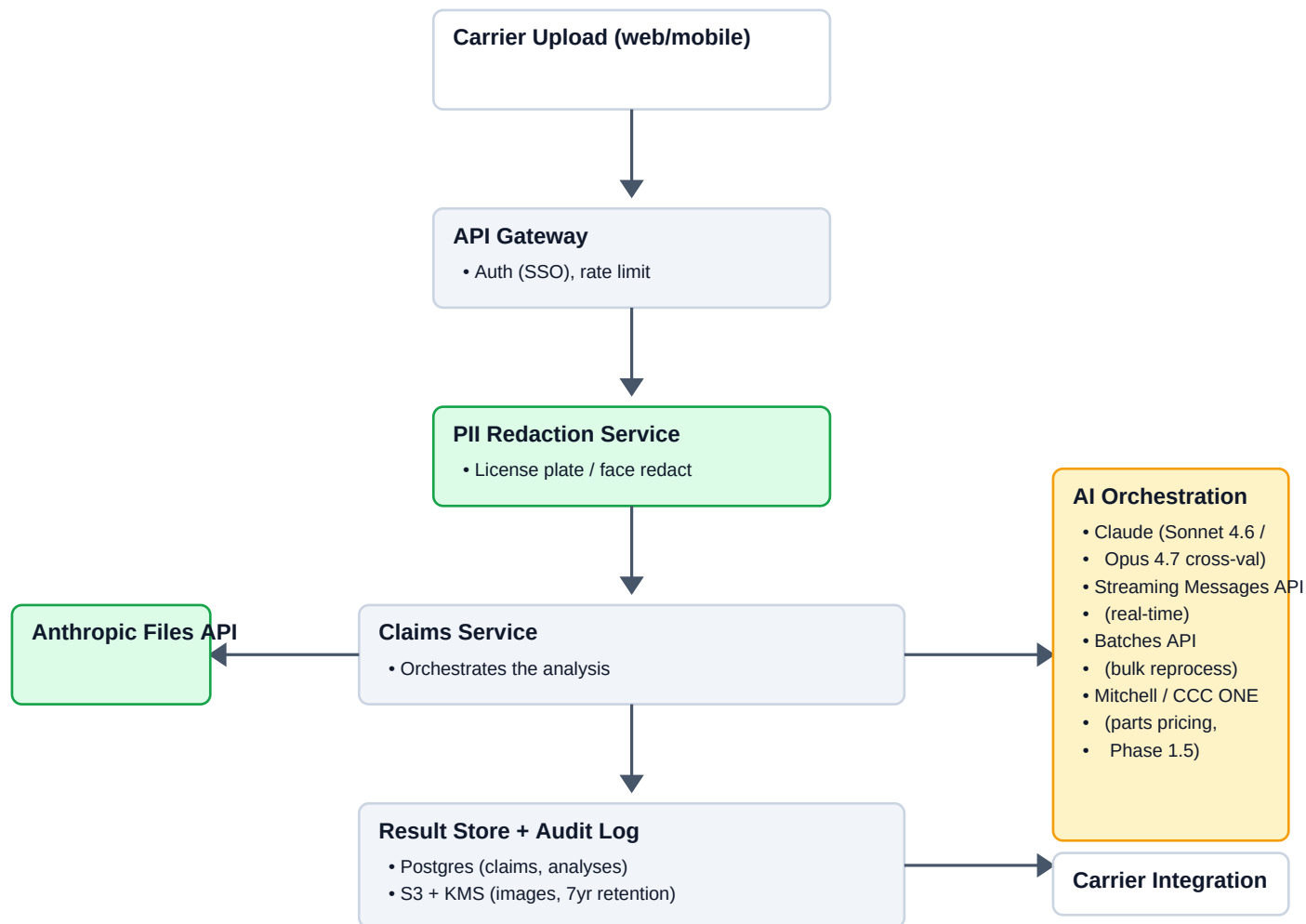


Figure 1. Production architecture. The streaming Messages API handles real-time per-claim analysis; the Batches API (Phase 1.5+) handles bulk reprocessing at 50% of synchronous-API cost.

AI / ML model and integration choices

Component	Choice	Justification
Primary vision-LLM	Claude Sonnet 4.6	Vision + structured outputs + adaptive thinking + streaming. Single-vendor reduces operational complexity.
Cost grounding (Phase 1)	In-prompt reference table	Industry-typical ranges by damage type × vehicle tier. No external dependency.
Cost grounding (Phase 1.5)	Mitchell / CCC ONE / Audatex API	Industry-standard parts-level pricing used by every major US carrier.

Component	Choice	Justification
Structured outputs	output_config.format JSON Schema	Server-enforced contract; downstream services consume typed JSON without parsing risk.
Image preprocessing	sharp (Node)	Resize, JPEG normalize, EXIF strip. Runs at the edge before image hits AI provider storage.
Confidence routing	Per-field 0–100 + typed Trigger objects (server-side)	Calibrated to adjuster workflow; thresholds tuned per carrier policy.
PII redaction	CV detection (license plate / face) + sharp blur	Required for compliance (state DMV regulations, GDPR-equivalent for international).
Cross-validation (>\$10k)	Run on Sonnet 4.6 + Opus 4.7; flag disagreement	Cheap insurance against single-model errors on high-stakes claims.
Self-consistency sampling	Re-run Sonnet 4.6 on the same image; flag variance	Triggered on low-confidence cases. Cheap variance signal that surfaces uncertain assessments adjusters should double-check.
VIN cross-reference (Phase 2)	NHTSA vPIC decode of dashboard VIN sticker	Catches AI make/model misidentification; fraud signal when claimed make/model ≠ VIN-decoded.
Bulk reprocessing (Phase 1.5+)	Anthropic Messages Batches API	Async batch endpoint for non-latency-sensitive workloads (model-upgrade reanalysis, fraud backfill, compliance audits, acquired-book reprocessing). Up to 100K requests / 24h at 50% of synchronous-API cost.
Interactive workflows (Phase 2+)	Anthropic Managed Agents API	Persistent stateful sessions for adjuster co-pilot and long-running fraud investigations. Anthropic runs agent loop and hosts per-session sandbox.
Persistence	Postgres + S3 (KMS-encrypted)	Standard. 7-year retention with Glacier archival per insurance regulation.

Integration points

- Carrier claims system: bidirectional. Outbound: AI assessment posted as claim attachment + structured fields. Inbound: claim status updates trigger reanalysis or archival.
- Carrier identity provider (SAML / OIDC): SSO for adjuster access to the review console.
- Carrier audit / SIEM: stream every read/write of claim data with actor, timestamp, purpose.
- Anthropic Claude API: Messages API (streaming) for real-time + Batches API for bulk + Files API for image storage.
- Mitchell / CCC ONE / Audatex (Phase 1.5): replace in-prompt reference table for parts-level pricing once carrier procures access.

Orchestration strategy: single-call spine + gated chaining

The Phase 0 prototype is a single `messages.create()` call. Production keeps the single-call architecture as the **default spine** for first-pass assessment — it handles the majority of claims correctly with one round-trip, one schema, and one transaction boundary. Chained model orchestration is layered on top **only as gated specializations** where a specific failure mode or cost dynamic justifies the extra inference. Six chained patterns ship across Phases 1–3:

Phase	Chain pattern	Trigger	What chains / Why
1	Cross-validation	high_usd > \$10k	Sonnet 4.6 + Opus 4.7 in parallel → deterministic comparison. Independent second opinion on high-stakes claims; flag disagreement for adjuster.

Phase	Chain pattern	Trigger	What chains / Why
1	Self-consistency sampling	Any field confidence < threshold	Re-run Sonnet 4.6 on same image → variance comparison on cost / severity / make-model. Variance signal that surfaces uncertain assessments adjusters should double-check.
1.5	Haiku triage tier	All claims (only ungated chain)	Haiku 4.5 image-quality + tier triage → escalate to Sonnet on pass, short-circuit on bad input. Cuts blended inference cost ~30% at scale; the only chain that reduces spend.
1.5	Parts-pricing fusion	Damage assessment complete	Sonnet damage description → Mitchell / CCC ONE / Audatex parts API → Sonnet cost grounding. Replaces in-prompt reference table with deterministic parts pricing.
2	Adjuster co-pilot	Adjuster opens claim console	Anthropic Managed Agents API — chained tool/model loop in a persistent session. Stateful multi-turn workflows: reanalyze, PDF/DOCX generation, iterative refinement.
3	Fraud pipeline	Similarity above threshold	Image-hash dedup → vector similarity → AI pairwise on top-K → structured fraud report. Multi-stage candidate narrowing makes the AI-pairwise step affordable.

Design rule. Every chain ships with: (a) a named failure mode it addresses, (b) a measured baseline error rate on that failure mode, (c) a trigger condition gating the chain to a subset of traffic, and (d) an A/B-validated metric move versus the single-call baseline. Chains that fail to earn their keep are removed in quarterly review.

Anti-patterns explicitly out of scope. The single-call assessment is *not* split into vehicle-ID → damage → cost stages — adaptive thinking already runs that reasoning internally inside one call, and splitting it adds latency, breaks cross-stage information flow, and creates coherence risk. Generic verifier stages without a named failure mode are also out of scope: two calls of the same model on the same input mostly agree with themselves, producing 2× cost for no real signal.

Deployment model

- Multi-tenant SaaS, with per-carrier data isolation at the database row level (PostgreSQL RLS) or schema level.
- Deployed in AWS us-east-1 with PrivateLink option for carriers requiring private connectivity.
- SOC 2 Type II in progress; HIPAA-equivalent controls for PII handling.

3. Milestones and Timeline

Engagement Timeline (Phase 0 → 4)



Figure 2. Phase timeline. Bar widths are proportional to phase duration. Phase 4 is ongoing.

Phase Milestone		Deliverables	Duration
0	Prototype (delivered)	Single-image upload + AI assessment. Local-run Next.js app + this SOW.	Complete
1	Pilot launch (production v1)	Hosted multi-tenant app, carrier SSO, persistence, audit log, PII redaction, claims-system integration (1 carrier).	8 weeks
1.5	Production cost integration + bulk reprocessing	Mitchell or CCC ONE replaces in-prompt reference for cost data. Operationalizes Anthropic Messages Batches API for non-latency-sensitive workloads (model-upgrade reanalysis, fraud backfill, compliance audits) at 50% of synchronous-API cost. Calibration data collection begins.	4 weeks
2	Multi-image flows + adjuster co-pilot	Multi-angle upload, side-by-side and AI-powered before/after comparison. Interactive adjuster co-pilot built on Anthropic's Managed Agents API — persistent stateful session per claim with PDF/DOCX generation.	6 weeks
3	Fraud-signal pipeline	Image-hash dedup, visual-similarity scoring against historical claims, AI pairwise comparison on top suspects. Long-running fraud investigations run as Managed Agents sessions.	8 weeks
4	National rollout + multi-carrier	Hardening, SLA tuning, observability, additional carrier onboarding, mobile-first UX refinements.	Ongoing

Phase 1 weekly breakdown

Week	Focus
1	Discovery: adjuster workflow shadowing, claim-volume profiling, ground-truth dataset assembly
2	Auth (SSO), persistence layer, deployment infrastructure
3	PII redaction service, audit log
4	Claims-system integration (carrier sandbox)
5	Adjuster review console (web UI), reanalysis flow
6	Calibration: tune confidence thresholds against ground-truth dataset
7	UAT with pilot adjuster cohort, bug fixes
8	Production cutover (limited cohort), runbook handoff to carrier IT

4. Risk Assessment

Risk	Likelihood	Impact	Mitigation
Cost estimate accuracy below adjuster expectations	Med	High	Phase 1.5 replaces reference table with Mitchell/CCC parts-level pricing. In Phase 1, cost is presented as a range with confidence; AI estimate framed as advisory. Confidence-routed escalation to human for low confidence.
AI misidentifies vehicle make/model on uncommon cars	Med	Med	Per-field confidence surfaced to adjuster. Adjuster can override. Calibration data from overrides feeds prompt tuning. VIN cross-reference via NHTSA vPIC (Phase 2) catches mismatches.
PII leakage (license plate, GPS in EXIF, face)	Low	High	Server-side EXIF stripping before any AI provider sees the image. Phase 1 adds CV-based plate/face redaction. SOC 2 controls and per-carrier data isolation.
Low-quality input photos (blur, dark, partial view)	High	Med	Image-quality scoring built into the AI pipeline; low quality caps confidence and triggers human review. UI surfaces specific quality issues so user can retake.
Adjuster distrust of AI / change-management resistance	High	High	Pilot launches as advisory tool only — no auto-decisions. Confidence indicators and source citations make every output auditable. Adjuster shadowing (Week 1) builds workflow alignment.
Anthropic API outage or rate-limit pressure during peak claim volume	Low	High	Multi-region failover, request queuing with SLA-bounded backoff, fallback to traditional desk-review queue. Business continuity runbook in place.
Cost overrun on AI inference at scale	Med	Med	Token-budget monitoring per claim. Image preprocessing caps input cost. Haiku 4.5 fallback for low-stakes claims (<\$2k) reduces per-claim cost ~5×. Caching + Batches API (50% off) for non-latency-sensitive workloads.
Regulatory: state-level disclosure requirements for AI in insurance decisions	Med	High	All Phase 1 outputs are advisory only — final decisions stay with licensed adjuster. UI clearly labels AI-generated content. Compliance review per state of operation.
Fraud actors exploit AI auto-handling on simple claims	Med	High	Phase 3 fraud-signal pipeline. Phase 1 mitigations: image-hash deduplication, sliding confidence thresholds for repeat claimants, sampling rate of human review on auto-handled claims.
Vendor lock-in (single-LLM dependency)	Low	Med	API contract for AI service is encapsulated in a single orchestration module. Phase 4 adds cross-vendor abstraction if needed. Open-source vision models evaluated as fallback for non-critical paths.

5. Success Metrics

KPIs are tracked from Day 1 of pilot. Targets are for the end of Phase 1 (8 weeks post-launch on the pilot cohort).

Primary KPIs

KPI	Baseline	Target	Why it matters
Time to first damage assessment	4–24h (adjuster desk)	< 5 min (AI-first pass)	Customer satisfaction; cycle-time reduction maps to NPS uplift
Adjuster touch-time per claim (simple claims)	~15 min	< 5 min	Adjuster capacity unlocked for complex claims
Cost-estimate accuracy (within $\pm 15\%$ of final)	N/A	$\geq 75\%$ on pilot cohort	Trust signal; unreliable estimates $\rightarrow$ adjusters won't lean on them
Auto-handle rate (simple claims, no escalation)	0%	30% of qualifying claims	Volume unlocked; revenue model justification
Adjuster override rate	N/A	< 25% of AI assessments	Calibration health; high override rate $\rightarrow$ retune prompt or thresholds

Quality and trust KPIs

KPI	Target
Confidence calibration (claimed vs. measured accuracy)	Within 10 percentage points
Severity-classification accuracy	$\geq 85\%$ agreement with adjuster
PII-redaction false-negative rate	< 0.1% (1 in 1000 images)
requires_human_review precision (when flagged, review warranted)	$\geq 80\%$
requires_human_review recall (claims that needed review were flagged)	$\geq 95\%$

Operational KPIs

KPI	Target
API availability (rolling 30 days)	$\geq 99.9\%$
p95 end-to-end analysis latency	< 60s
p99 end-to-end analysis latency	< 90s
Cost per analyzed claim (compute + AI)	< \$0.50
Audit-log completeness	100% (compliance)
Bulk reprocessing SLA (Batches API runs, Phase 1.5+)	$\geq 95\%$ within 24h
Bulk-vs-realtime cost delta (validation of 50% savings)	$\geq 45\%$ measured savings on Batches workloads

Business KPIs (Phase 1.5+ as data accumulates)

- Loss-cost reduction on pilot cohort vs. control cohort: target 3–5% via faster cycle-time and more consistent estimates.
- Customer NPS on claim handling: target +10 points on pilot vs. baseline.

- Adjuster satisfaction with the AI-assist workflow: target 4.0+ on a 5-point scale by end of Phase 1.

Reporting cadence

Cadence	Reports
Weekly	Operational KPIs + adjuster override rate (during Phase 1)
Bi-weekly	Quality KPIs (confidence calibration, severity accuracy)
Monthly	Business KPIs + pilot vs. control comparison
Quarterly	Executive review with carrier leadership

Appendix A — Sign-off

Role	Name	Date
Carrier Project Sponsor		
Carrier Claims Operations Lead		
Solutions Engineering Lead		
Implementation Manager		